

# Predicting NFL Team Records

## A Game-by-Game Win Projection System for the 2025 Season

*Chicago Bears Football Analytics Interview Project*

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### Abstract

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This project builds a system that produces an updated projected final win-loss record for all 32 NFL teams before every game of the 2025 season. The work uses only Pro Football Reference data, sourced through `nfl_data_py`, across the 2018 to 2025 seasons. I engineered 42 pre-game features, including a custom offensive and defensive ELO system driven by net yards per play, and ran a five-model suite on two parallel tracks: one using on-field statistics alone, and one that adds the Vegas spread and total as a benchmark. The headline result is that a neural network trained on on-field statistics alone reaches 65.6% game-level accuracy on the 2025 holdout, matching the Vegas straight-up baseline of 65.6%. Adding market data on top of the statistical model produced no meaningful lift, which says the publicly available efficiency metrics already carry most of the recoverable signal. The projections feed a deployed application and a post-model salary cap analysis that I tie directly to how the Bears have approached their rebuild.

### The Problem and the Bar

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The deliverable is simple to state and hard to do well. Before each game of the 2025 season, generate a projected final record for every team, and update it as results come in. The harder question underneath it is the one worth answering: can on-field statistics predict NFL outcomes as accurately as the betting market?

The market is the right bar because it is honest and public. A closing spread already prices in injuries, weather, travel, rest, and private scouting that no public dataset contains. On the 2025 test set, the Vegas spread picks the straight-up winner 65.6% of the time at the game level. If a model built only on box score and tracking efficiency can reach that number, then those efficiency metrics are doing real work, and a front office can act on them without paying for market information. That framing is the whole point of the two-track design described below.

### Project Timeline

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The project moved through a few clear phases, and the version history matters because most of the work was in getting the validation honest rather than chasing accuracy.

- **Data sourcing.** My first approach scraped Pro Football Reference directly. It was rate-limited and broke whenever the page HTML changed, so I moved to `nfl_data_py`, which exposes the same PFR advanced metrics through a clean, versioned API. That preserved PFR fidelity while removing the scraping fragility.
- **Feature build and ELO.** I reshaped the schedule into two rows per game, joined the advanced passing, rushing, and defensive tables, and built the rolling pre-game features and the split offensive and defensive ELO ratings.
- **Model suite.** I expanded from a single classifier to five model types, then added a point-differential regression track and the Vegas-augmented track for benchmarking.

- **Validation hardening.** Successive versions fixed real leakage problems: early stopping that was peeking at the test year, playoff-stakes flags that were built from final standings, and rolling fills that used future games. Each fix is documented in a leakage audit table inside the notebook.
- **Projection engine and app.** Finally I wrapped the win probabilities into a season projection with Monte Carlo intervals and shipped a deployed Streamlit application so the output is usable, not just a notebook.

## Data

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All model data comes from Pro Football Reference through `nfl_data_py` and covers the 2018 to 2025 regular seasons. I start at 2018 because that is the earliest year PFR publishes the advanced tracking stats this project leans on, including pressure rate, yards after contact, and coverage metrics. Running the model on years before that would mean training on an incomplete feature set.

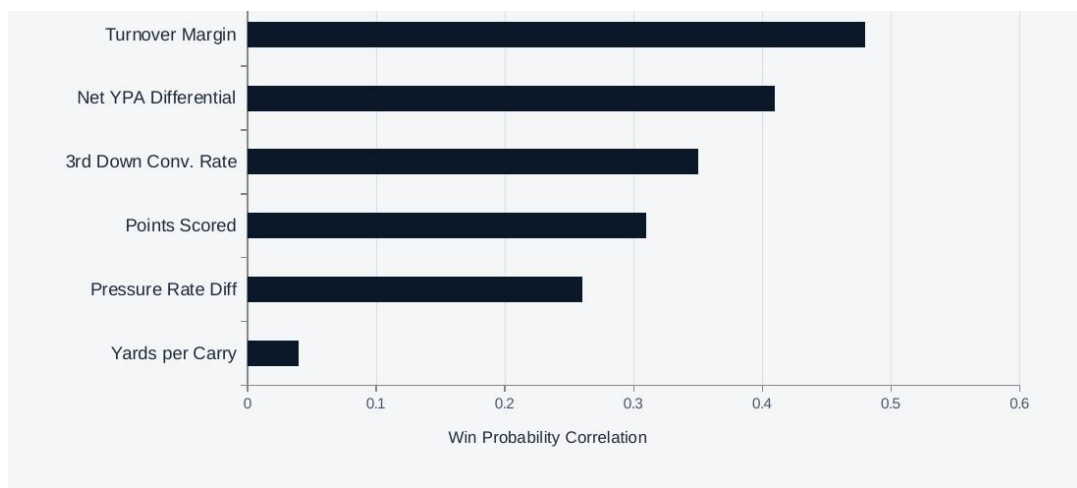
The base table holds 4,254 game rows. Each game appears twice, once from each team's perspective, which makes it natural to express every feature as a team-versus-opponent quantity. The target is the binary win for classification and the point differential for regression. Situational stats that PFR does not publish at the game level, such as third down conversion rate, were rebuilt from play-by-play data. The training window is 2018 to 2024, weeks 1 to 17. The test set is the 2025 season, weeks 1 to 17, with week 18 held out separately because clinched and eliminated teams behave differently in a meaningless game.

## Methods

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### Exploratory data analysis

Before modeling, I checked which game-level quantities actually move win probability. Turnover margin and net yards per play differential are the two strongest single signals, followed by third down conversion rate and points scored. Yards per carry, the stat fans talk about most, is nearly flat. That pattern shaped the feature set: efficiency and ball security over raw volume.



*Figure 1. Game-level correlation with winning, 2018 to 2024. Turnover margin and net yards per play differential lead; yards per carry barely registers.*

### Feature engineering

Every feature is strictly pre-game. Team efficiency stats are turned into expanding rolling averages that are lagged by one game, with a four-game exponentially weighted version to capture recent form. I adjust the main efficiency metrics for strength of schedule using the pre-game rolling quality of opponents faced, and I seed week 1 of each season from the prior season regressed toward the league mean. Context features include home field, rest and net rest differential, week of season, division game, and the playoff-stakes flags. The final base set is 42 features across seven groups; the Vegas set adds the spread and total for 44.

### **Offensive and defensive ELO**

Standard team ELO treats a team as one unit. Football has two distinct contests in every game, so I split it into an offensive ELO and a defensive ELO for each team. The update signal is net yards per play, meaning passing and rushing combined and net of sack yardage, compared against the season-to-date median. An offense that clears the median earns offensive ELO and costs the opposing defense the same amount, which keeps the two ratings coupled. Ratings regress one third of the way back to 1500 between seasons. In 2025 the offensive ratings spanned roughly 1340 to 1644 and the defensive ratings 1383 to 1636. Splitting the rating matters most early in the year, when a 17-game sample is thin and each unit's rating is more stable than a combined one. ELO turned out to be the single strongest input to both model tracks.

### **A two-track model suite**

Each track runs the same five model types: XGBoost, Random Forest, Logistic Regression, Ridge, and a neural network (a multilayer perceptron). I run them as direct win-loss classifiers and, separately, as point-differential regressors whose predicted margin is mapped back to a win. Hyperparameters for every model are chosen with walk-forward cross-validation, training on seasons up to year  $S$  and validating on year  $S+1$ , so the tuning never sees future data. The non-Vegas track answers whether on-field stats explain wins. The Vegas track adds the spread and total purely to measure how much the market knows that the statistics do not.

### **Guarding against leakage**

Because the accuracy bar is the market, an inflated number would be worse than useless. I ran an explicit leakage audit covering rolling features, the strength-of-schedule adjustment, the ELO snapshot, the playoff-stakes flags, train-test imputation, and the target columns, and fixed each issue I found. The most important fix was XGBoost early stopping, which in an earlier version was guided by the 2025 test loss. I moved it to an out-of-time slice from 2023 and 2024 so the test season is never seen during training. Accuracy is always reported at the game level, one independent prediction per game, so it is directly comparable to the Vegas spread.

### **From game probabilities to season records**

The projected win total for a team is its actual wins so far plus the sum of its model win probabilities over the remaining schedule. To put honest uncertainty on that number, I run 5,000 Monte Carlo simulations of each team's remaining games, drawing a Bernoulli outcome per game from the model probability, and report the 10th to 90th percentile band. Wide bands flag teams with many close games; narrow bands flag decisive mismatches. The full week-by-week projection table, the intervals, and the schedule view all live in the deployed app at [the project Streamlit link](#).

## **Results**

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The central finding is in the table below. On the 2025 holdout, the neural network trained on on-field statistics alone reaches 65.6% game-level accuracy, matching the Vegas straight-up baseline of 65.6%. The tree models and linear models trail in the low 60s. When I add the spread and total to the models, the best Vegas-augmented model, a Ridge classifier at 65.2%, lands just below both the non-Vegas neural network and the raw Vegas baseline. In other words, the statistical model already extracts most of the available signal, and the market adds little on top of it.

| Model                        | No-Vegas     | Train        | Vegas track  | Train        |
|------------------------------|--------------|--------------|--------------|--------------|
| XGBoost                      | 62.1%        | 75.7%        | 63.3%        | 77.7%        |
| Random Forest                | 61.7%        | 68.7%        | 64.5%        | 76.7%        |
| Logistic Regression          | 60.9%        | 63.1%        | 64.8%        | 67.5%        |
| Ridge Classifier             | 60.6%        | 63.2%        | 65.2%        | 67.5%        |
| <b>Neural Network (MLP)</b>  | <b>65.6%</b> | <b>62.7%</b> | <b>63.7%</b> | <b>71.4%</b> |
| <b>Vegas spread baseline</b> | <b>65.6%</b> | n/a          | <b>65.6%</b> | n/a          |

Table 1. Game-level accuracy on the 2025 holdout (weeks 1 to 17). The non-Vegas neural network matches the Vegas baseline; adding market data does not improve on it.

Two diagnostics give me confidence the headline number is real rather than lucky. The model is well calibrated: when it says 60%, teams win at about 60%, with deviation only at the extreme bins where there are few games. And the feature attributions point at football-sensible drivers rather than noise.

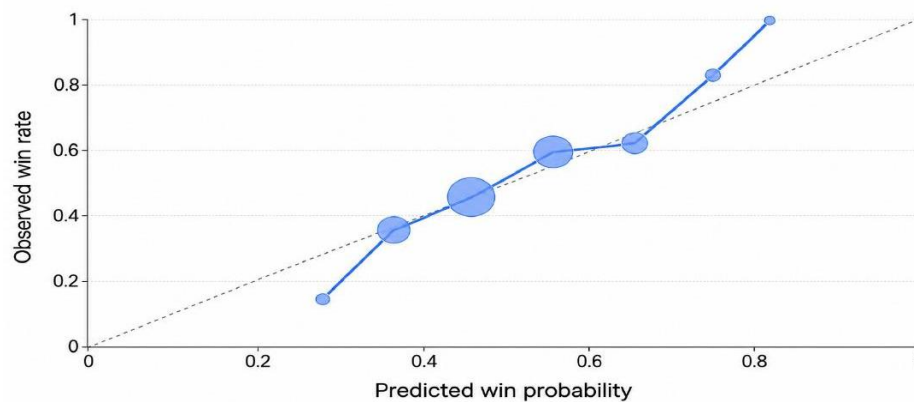
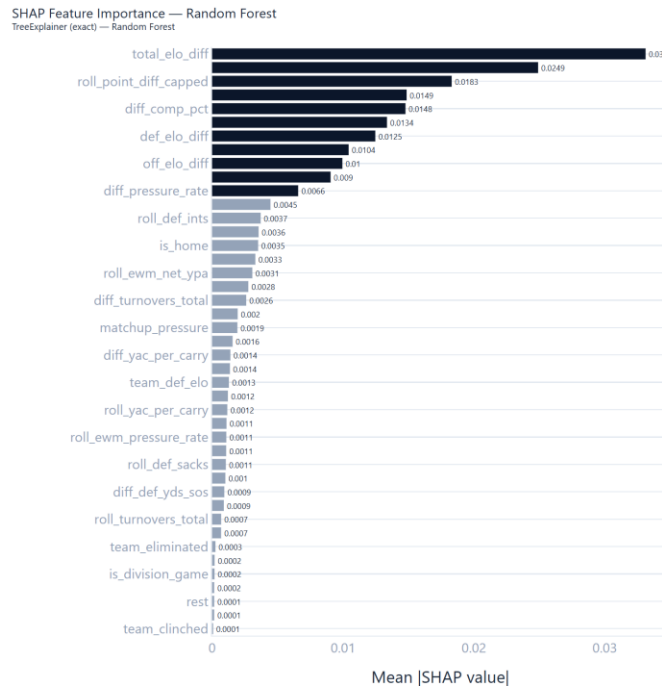
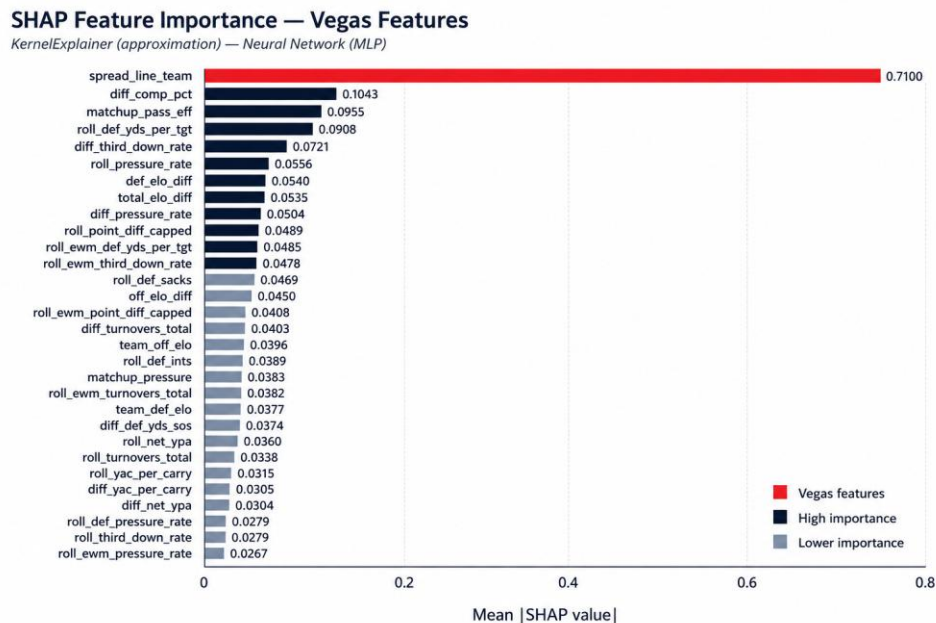


Figure 2. Calibration on the 2025 holdout. Predicted win probability tracks observed win rate closely; bubble size is games per bin.



**Figure 3.** Feature attribution for the Random Forest, the most interpretable model in the suite. The split ELO terms and rolling efficiency lead; raw volume stats sit near the bottom.

The ELO differential terms dominate the attributions in both tracks, which confirms the EDA story: efficiency and matchup quality carry the prediction. In the Vegas track the spread predictably dominates, but the on-field features that survive next to it, completion percentage, matchup pass efficiency, and defensive yards per target, are exactly the ones the market does not fully price.



**Figure 4.** Vegas-track attribution. The spread is roughly six times any other feature, yet specific efficiency metrics retain independent value.

On the actual deliverable, projecting final records, the preseason win-total error is 2.32 wins for the non-Vegas model and 2.00 wins for the Vegas model. Both are competitive before a single game is played, and the errors

shrink as the season fills in. Tracked week by week, the neural network stays close to the Vegas baseline through the year and only slips in week 18, the meaningless-game week the model was never meant to handle.

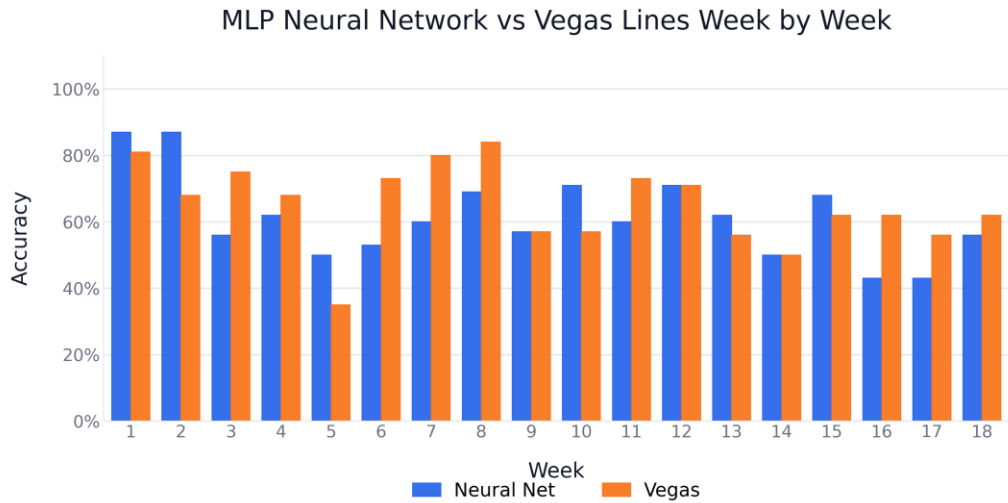


Figure 5. Week-by-week game-level accuracy, neural network versus Vegas. The two run nearly even across the season.

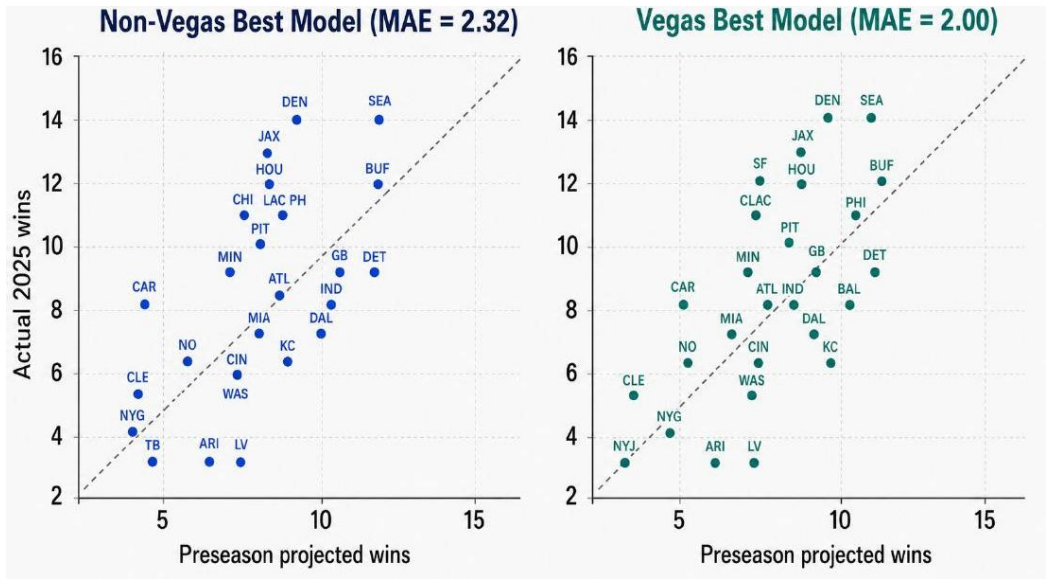


Figure 6. Preseason projected wins versus actual 2025 wins. Points above the diagonal are teams that beat their projection. Chicago is one of them.

### Implications for Bears Team Building

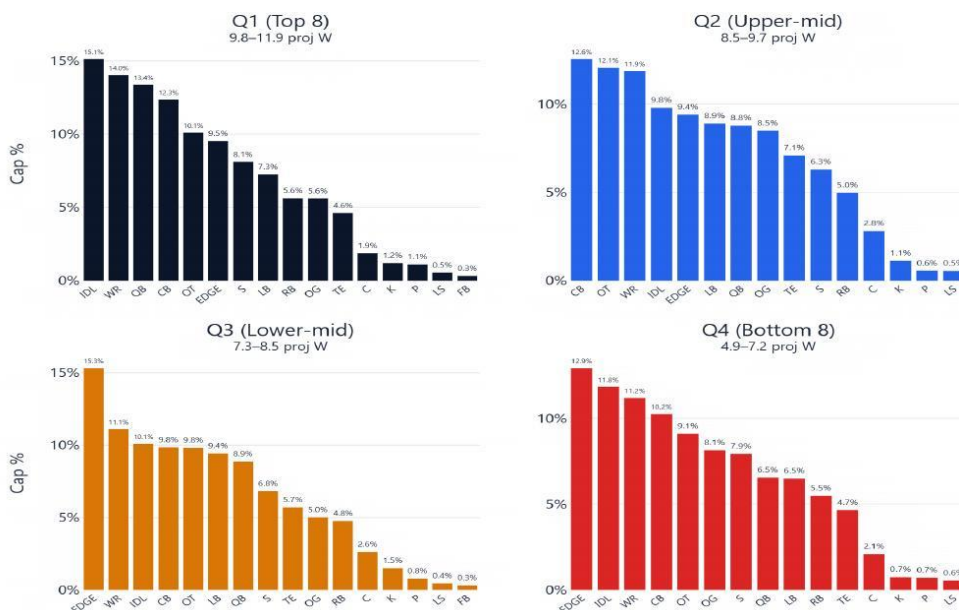
A win model is only interesting to a front office if it points at decisions. Three implications fall out of this work, and each lines up with how the Bears have actually operated.

First, the model under-projected the rebuild. The non-Vegas model opened Chicago around eight wins, anchored on inputs trained through 2024, and the team finished well above that line in 2025. A model that leans on prior-season efficiency will always lag a roster that improves faster than its track record. That gap is the quantitative version of the discipline I admire in this rebuild: moving off bad contracts, drafting high in

consecutive years, and adding premium-position talent in Caleb Williams, Rome Odunze, and Colston Loveland changes a team faster than a backward-looking model expects.

Second, the stats that drive the model are the stats worth paying for. Turnover margin, net yards per play differential, and pressure rate sit at the top of both the EDA and the feature attributions, while raw volume barely moves the needle. That argues for valuing players by their effect on efficiency and ball security rather than on counting stats, which is the same logic behind trading DJ Moore for a second round pick when the surplus value said to. It also reinforces a defense, like the one Dennis Allen runs, that generates pressure and limits explosive plays while staying structurally sound.

Third, the cap analysis. As a post-model exercise (it uses Over The Cap data and touches none of the predictions), I ranked teams into quadrants by projected wins and looked at how each group spends. The top-eight projected teams concentrate cap dollars on premium positions, with interior defensive line, receiver, quarterback, and cornerback near the top, while the bottom group skews its spending differently. This is descriptive rather than prescriptive, and correlation is not causation, but it is the kind of pattern a front office can use to sanity-check allocation. It maps cleanly onto a Bears approach that prioritizes premium positions and is willing to be uncomfortable when the numbers say to be.



**Figure 7.** Positional cap allocation by projected-win quadrant (post-model, Over The Cap data). Higher-projected teams concentrate spend on premium positions.

This system also sits next to my Defensive Personnel Optimizer, a Bayesian hierarchical model that prescribes base, nickel, or dime by expected EPA. Together they reflect how I want to contribute: build the data pipeline honestly, validate against a real bar, and turn the output into something a coach or a cap manager can actually use on a given day.

## Conclusion and Future Work

The result I trust most is the modest one stated plainly: a model built on public, on-field statistics can match the market at picking NFL winners, and once it does, the market adds almost nothing on top. That is a useful thing for a front office to know, because it means the efficiency metrics are a defensible basis for decisions rather than a pale shadow of the spread.

There is clear room to grow. Red zone efficiency and EPA per play are gold-standard signals that PFR does not expose at the game level, and adding them is the obvious next lift. Injury indicators would close part of the gap to the market. A Bayesian or Kalman-filter ELO would put honest uncertainty on the ratings themselves, and the clinch and elimination logic deserves exact playoff math rather than a wins-and-losses proxy. None of these change the core conclusion. They sharpen a framework that already does the job it was asked to do.

## References

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Pro Football Reference ([pro-football-reference.com](https://pro-football-reference.com)). All model data.

nflverse project and nfl\_data\_py ([nflverse.com](https://nflverse.com)). Data pipeline and hosting.

FiveThirtyEight NFL ELO methodology. Season carryover and regression approach.

Over The Cap ([overthecap.com](https://overthecap.com)). 2025 positional cap spending, used only in the post-model supplementary analysis.